

# Sepehr Kahvand

M.Eng in Mechanical Engineering | Mechanical Designer | CSWA | GD&T & Tolerance Design  
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## SUMMARY

Mechanical Engineering Professional with 10 years of progressive experience supporting automotive and industrial manufacturing through mechanical design, production tooling, quality, and process improvement. Proven ability to collaborate with Engineering, Manufacturing, Quality, and Production teams to resolve technical challenges, improve processes, and support successful engineering and manufacturing initiatives. Strong background in GD&T, tolerance analysis, DFM, manufacturing processes, production tooling, dimensional inspection, and ISO 9001 / IATF 16949 quality systems. Demonstrated record of delivering measurable improvements, including increased production efficiency, reduced defects and rework, and significant cost savings. Combines hands-on manufacturing experience with strong technical, analytical, and problem-solving skills to drive continuous improvement and support production objectives.

## RELATED PROFESSIONAL EXPERIENCES

### Linamar Corporation (*Comtech Mfg.*)

#### ❖ **Quality & Metallurgy Lab Technician** – Permanent Full-time

Aug 2026 – Present

- Perform **precise dimensional inspections** of automotive powertrain components using **CMMs, Vernier calipers, micrometers, shadow graphs**, and other precision gauges to verify engineering specifications and GD&T requirements
- Measure surface finish and roughness using **surface roughness testers**, and conduct **volumetric testing** to verify **cylinder head chamber volume**
- Operate **advanced inspection equipment**, including **gear analyzers** and **contour graphs**, to **assess profiles and surface contours of precision components** such as intake and exhaust valve seats
- Conduct **metallurgical laboratory testing**, including **hardness testing, microstructure evaluation**, and material analysis, to support quality control and product validation
- Document inspection results, identify non-conformances, and communicate findings to Engineering, Manufacturing, and Quality teams while supporting root cause analysis and corrective actions
- Collaborate with cross-functional teams to maintain high product quality and support continuous improvement and production efficiency

#### ❖ **Computer Numerical Control Machine Operator** – Permanent Full-time

Aug 2025 – July 2026

- Operate **CNC lathe machines** to manufacture transmission cover plates in a **fast-paced, high-volume production environment**
- Perform **tool insert changes, machine adjustments, and offsets** to ensure parts meet nominal dimensions and quality specifications
- Inspect machined parts every 2 hours using **Mueller gauge, height gauge, and Go/No-Go gauges**, completing **Operator Data Sheets** for QA documentation
- Identify and report **non-conforming products**, tag defective parts, and halt production until approval from supervisors or team leaders
- **Improved production output by 47% by analyzing the automation workflow and collaborating with the program manager and automation supervisor to optimize the logic of the automated system**

### Faraz Sanat Engineering Co.

#### ❖ **Mechanical Designer** – Permanent Full-time

Apr 2020 – Apr 2024

- Designed mechanical subassemblies, custom jigs, dies, fixtures and production tooling using **SolidWorks**
- Performed tolerance stack-ups and GD&T analysis to ensure manufacturability and assembly reliability
- Generated and maintained accurate **bills of materials (BOMs)** and drawing packages for internal and supplier use
- **Led tooling and fixture design improvements through detailed 3D modeling and GD&T application, collaborating with engineers and production teams to achieve 2.7% production line efficiency increase, reduced assembly time, and lower manufacturing errors—delivering estimated annual savings of \$185,000**

## Niroo Motor Group

❖ **Quality Assurance Specialist / Quality Control Inspector – Motorcycle Assembly Division – Permanent Full-time**  
*Jan 2015 – Mar 2020*

- Led investigations into assembly line defects such as misalignment, paint damage, wiring errors, and loose fasteners, using root cause analysis tools (5 Whys, Fishbone)
- Issued and tracked Corrective and Preventive Actions (CAPA) for recurring issues, collaborating with assembly, logistics, and supplier teams
- Supported compliance with ISO 9001 and IATF 16949 by maintaining updated records, calibration logs, and audit-ready documentation
- Reported key quality metrics (defect rate per station, rework frequency, customer complaint trends) to QA Manager with recommendations for long-term improvements
- **Reduced rework and defect rates on the final assembly line by over 20% within two years through process optimization, better operator training, and systematic issue tracking**

## EDUCATION

- Master of Engineering: Mechanical Engineering May 2024 – Apr 2025  
University of Guelph, Canada | CGPA: 92/100 | **Graduated with Top Academic Standing**
- Bachelor of Science: Mechanical Engineering (Solid Mechanics and Design)  
University of Guilan, Iran

## CERTIFICATIONS

- **Certified SOLIDWORKS Design Associate**  
*Dassault Systèmes | Completed: April 12, 2026*
- **Geometric Dimensioning & Tolerancing: Advanced concepts**  
*Udemy | Completed: April 5, 2026*
- **Worker Health & Safety Awareness – Ontario Regulation 297/13**  
*University of Guelph | Completed: January 9, 2025*
- **WHMIS 2024 – Workplace Hazardous Materials Information System**  
*University of Guelph | Completed: June 26, 2024*
- **Advanced Python Programming and Object-Oriented Thinking Course**  
*Quera College | Completed: December 2022*
- **Internal Quality Auditor – ISO 9001:2015 (Manufacturing Focus)**  
*GLIS Training Solutions | Completed: 2020*

## SKILLS

### Technical Skills:

- |                                   |                       |                     |                            |
|-----------------------------------|-----------------------|---------------------|----------------------------|
| • Computer-Aided Design (CAD)     | • SolidWorks          | • AutoCAD           | • FMEA/FMECA               |
| • BOM Documentation               | • GD&T (ASME Y14.5)   | • Tolerance Design  | • Test Plan Development    |
| • Autodesk Inventor               | • Reliability Testing | • CNC lathe machine | • MATLAB/Simulink          |
| • ISO 9001:2015   IATF 16949:2016 | • Python              | • Microsoft Excel   | • FEA (ANSYS & SolidWorks) |

### Professional Competencies:

- |                   |                       |                       |                   |
|-------------------|-----------------------|-----------------------|-------------------|
| • Teamwork        | • Communication       | • Adaptability        | • Time Management |
| • Problem Solving | • Analytical Thinking | • Attention to Detail |                   |

## REFERENCES

- **References available upon request.**